Analytics & Big Data

Session 0: Course Logistics

Prof. Dr. Gerit Wagner

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Welcome

Prof. Dr. Gerit Wagner

Academic background

- Universität Regensburg: Doctorate
- HEC Montréal: Postdoctoral Fellowship
- Otto-Friedrich-Universität Bamberg: Assistant Professor
- Frankfurt School of Finance & Management: Full Professor

Research interests

- Open, agentic, and boundary-spanning work
- AI-supported knowledge synthesis

Teaching interests

- Programming and software engineering
- Analytics and big data
- IT security
- Digital knowledge-intensive and platform-based work
- Literature review methods

About you

Please answer the three questions. Then choose **one** optional area where you would like to share a bit more.

1. Have you worked in an industry role or completed an internship with an analytics focus?

Optional:

- What was something that felt **very useful** in your analytics work?
- Or where would you have liked to be **better prepared** for a data-related task?

2. Which analytics tools or programming languages have you used?

Optional:

- What was a particularly interesting analytics **challenge** you encountered?
- Or a tool you found especially powerful or **surprising**?

Analytics & Big Data

Analytics & Big Data in the curriculum

1*	Principles of Management	Financial Accounting	Introductionto Programming	Mathematics	Analytical Thinking & Critical Reasoning
2*	Principles of Finance	Microeconomy	Statistics& Probability	Macro-economics	Marketing
3*	Operations Management	Managerial Accounting	Markets, Incentives & Ethical Management	Applied Research Methods	BusinessLaw
4	Corporate Finance	Strategy & Organisation	Concentration		
5	SemesterAbroad &Internship Abroad				
6	Elective	Academic Skills Module	Concentration		
7	Elective	Elective	Elective	Thesis	

Prerequisite: Completed the “Introduction to Programming” module.

What exactly is "analytics"?

Business analytics refers to the computer-supported examination of data using mathematical models to drive decisions and actions within business situations (based on Davenport & Harris 2007).

Why Analytics (and This Course) Matter

TODO: https://en.wikipedia.org/wiki/Princeton_Newport_Partners (black-scholes)

💰 Finance — Algorithmic Trading at Renaissance Technologies

- Investment decisions driven almost entirely by statistical models
- Tiny predictive advantages scale across millions of trades
- Data replaces intuition in high-stakes financial markets

🎬 Entertainment — Personalization at Scale at Netflix

- Recommendation systems shape what millions watch
- Continuous A/B testing and model refinement
- Data directly drives engagement and revenue

⚾ Sports — The Moneyball Revolution at Oakland Athletics

- Advanced metrics replaced traditional scouting intuition
- Data-driven player selection with limited budget
- Competitive advantage through better measurement

Course Architecture

Learning objectives

Knowledge

- **Differentiate** between descriptive, predictive, and prescriptive analytics approaches.
- **Describe** architectures for data warehousing and big data processing, with reference to the underlying data modeling concepts.
- **Explain** fundamental concepts of data analytics, and its role in data-driven organizations.

Skills

- **Select** appropriate analytical methods for transactional or non-transactional data, and for descriptive, predictive, and prescriptive analysis tasks.
- **Apply** analytical methods to real-world datasets, including structuring, transforming, and visualizing data. This also involves training models, evaluating performance, and interpreting analytical results for decision-making.
- **Implement** analytical procedures in Python, using standard data science libraries (pandas, scikit-learn, matplotlib).

Competence

- **Design** analytics solutions aligned with business goals and governance principles.
- **Integrate** analytical technologies into organizational decision-making processes
- **Assess** the operational and strategic implications of data analytics in organizations.

What This Course Does *Not* Cover

- **Engineering of production-grade analytical systems** (software engineering, database design and optimization, DevOps/MLOps pipelines, scalable application development, distributed systems implementation)
- **Statistical theory** (inferential statistics, proofs/derivations)
- **Autonomous / agentic decision systems** (robots/agents acting on decisions, reinforcement learning, LLM engineering)

> **Note**
>
> **Interfaces exist:** We will *touch* these areas where needed, but our focus is on **analytical reasoning** and **prototype implementations** (e.g., notebook-based workflows) for real-world datasets.

Course logistics

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Workload: 150h total (44h in class, remaining: self-study and group project)

Assessment

- Exam: 60 minutes, 60 points
- Group project: written paper (40 pages), 60 points (end of module)

Sessions:

- See overview in Canvas

Contact:

- g.wagner@fs.de
- Office hours: booking page

Individual circumstances

If you have family responsibilities, religious holidays, health-related matters, or other individual circumstances that may affect your participation or performance, please reach out early. We will work together to find a fair and workable solution.

Materials

Slides and materials

- Presentation slides will be made available for download.
- You are expected to take complementary notes and read the recommended literature.
- You can contribute directly to the teaching materials by submitting an issue ♻️ or suggesting edits 🛠️.
- Literature and complementary materials will be listed at the end of each lecture.
- Reading of complementary materials depends on your interest and ambition.

Selected literature

- Part 1: Chen et al. (2012), Martínez-Plumed et al. (2019), Sharda et al. (2018)
- Part 2: Raschka (2015), Schutt & O'Neil (2014), Vaisman & Zimányi (2016)
- Part 3: Bengfort et al. (2018). Schmarzo (2016)
- Part 4: Davenport (2006), Vidgen et al. (2017)

You may also be interested in ...

Bachelor's theses

- See information on the process and open topics

TODO

TODO: update thesis topics

References

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Chen, H., Chiang, R. H. L., & Storey, V. C. (2012). Business intelligence and analytics: From big data to big impact. *MIS Quarterly*, *36*(4), 1165–1188. https://www.jstor.org/stable/41703503

Davenport, T. H. (2006). Competing on analytics. *Harvard Business Review*, *84*(1), 98–107. https://cs.brown.edu/courses/cs295-11/competing.pdf

Martínez-Plumed, F., Contreras-Ochando, L., Ferri, C., Hernández-Orallo, J., Kull, M., Lachiche, N., Ramírez-Quintana, M. J., & Flach, P. (2019). CRISP-DM twenty years later: From data mining processes to data science trajectories. *IEEE Transactions on Knowledge and Data Engineering*, *33*(8), 3048–3061. https://ieeexplore.ieee.org/abstract/document/8943998

Raschka, S. (2015). *Python machine learning*. Packt Publishing. https://www.packtpub.com/en-us/product/python-machine-learning-9781783555130

Schmarzo, B. (2016). *Big data MBA*. Wiley. https://doi.org/10.1002/9781119238881

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Vaisman, A., & Zimányi, E. (2016). *Data warehouse systems: Design and implementation*. Springer. https://doi.org/10.1007/978-3-662-65167-4

Vidgen, R., Shaw, S., & Grant, D. B. (2017). Management challenges in creating value from business analytics. *European Journal of Operational Research*, *261*(2), 626–639. https://www.sciencedirect.com/science/article/pii/S0377221717301455